Satyam Chaturvedi

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PROFESSIONAL SUMMARY

Final year B. Tech (CSE) student, Software Engineer with an applied focus on Applied AI & Machine Learning systems for working on high impact agentic AI problems in applied, research and product directions. Building core machine learning, AI solutions with the help of LLM or AI infrastructure.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, JavaScript

Agentic AI & Generative AI: LangGraph, LangChain, LlamaIndex, Google ADK, RAG Pipeline, Prompt Engineering, Embeddings, OpenAI API, Gemini API, Transformers, PyTorch, TensorFlow, Scikit-learn

Web & API Development: FastAPI, Flask, RESTful APIs, Async Programming, Streamlit, Next.js

Databases & Data: PostgreSQL, MySQL, MongoDB, PGVector, SQLAlchemy, Redis, Kafka

Cloud & MLOps: AWS, GCP, Azure, Docker, Kubernetes, Git, GitHub, Linux, CI/CD

AI Platforms & Engineering: Azure OpenAI, Google Vertex AI, AWS Bedrock, Claude Code, Codex

EXPERIENCE

Software Development and Training Intern - Black Orange Talent Pvt. Ltd.

Delhi, India | July 2026 – Oct 2026

- Building a Learning Management System (LMS) portal and contributing AI and RAG support to a system serving 100+ organizations and 50K+ students across PAN India.
- Developed a GenAI based Web Applications for the company integrating TTS and STT services.
- Delivered STEM education and technical training to 1K+ students across 20+ organizations across Delhi NCR, effectively communicating technical concepts to diverse K-12 audiences.

PROJECTS

CFOBuddy - Agentic AI Co-Pilot | *Repo* / *Live App* | ***Tech Stack:*** *LangGraph, LangChain, LlamaIndex, PGVector, FastAPI, pgVector*

- Built an **agentic AI co-pilot** to streamline financial analysis for internal users, designing *a **3-tier workflow with 4 specialized agents using LangGraph***. Replaced LLM-based routing with a lightweight semantic/keyword fallback, reducing response latency to **200ms**.
- Implemented *a **Text-to-SQL** **engine*** with schema introspection, few-shot prompting, and structured error recovery for natural-language analytics over dynamically uploaded datasets on PGVector, achieving ***92% response accuracy***.
- Designed a ***hybrid RAG pipeline*** combining BM25 sparse search and PGVector semantic search with optimized chunking and cross-encoder reranking, achieving a **9% error rate** on RAG queries.

Radiology Reporting Assistant - Template-Aware Clinical LLM | *Repo* / *Model* / ***Tech Stack:*** *Python, PyTorch, Hugging Face Transformers, Qwen2.5-3B-Instruct, PEFT, QLoRA, TRL, Gradio*

- Fine-tuned using **QLoRA** and **supervised fine-tuning (SFT)** to transform radiology dictations into structured, template-aligned reports.
- Developed a **comparative evaluation pipeline** and benchmarked against a copy-template baseline across **95 test cases**, achieving a **54.99% reduction in mean Radiology Edit Score (RES)**.
- Achieved **56.27% lower findings scores** and **53.09% lower impression scores**, the fine-tuned model recorded a lower RES than the baseline on **81 of 95 cases (85.3%)**.
- Analyzed model failures and performance across **imaging modalities and body parts** to identify underperforming cases and guide further evaluation.

AssessQ - Assessment Recommendation Engine | *Repo* / *Live App* | ***Tech Stack:*** *Python, ChromaDB, Sentence-Transformers, LangChain, Streamlit*

- Built and deployed a **hybrid retrieval pipeline** combining BM25 and dense embeddings with **Reciprocal Rank Fusion (RRF)**, evaluated **4 retrieval configurations across 30 queries**, achieving ***0.90 Recall@5*** and improving retrieval performance over the BM25 baseline.

EDUCATION

Bachelor of Technology in Computer Science and Engineering Expected: 2027

ABES Institute of Technology, Ghaziabad, India

Relevant Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Theory of Computation, Compiler Design, Computer Networks, Software Engineering

TECHNICAL EXPERIENCE & ACTIVITIES

- Selected as an Intern for the **Infosys Springboard Virtual Internship 2026**; completed the AWS AI & ML Scholars (AI Practitioner) program, building a productivity app on AWS PartyRock.
- Completed the **Google Cloud Gen AI Academy APAC Edition (Cohort 1)** and the **Google × Kaggle 5-Day Agent Intensive course**, alongside multiple hackathons.

ACHIEVEMENTS

- Solved 1000+ Data Structures & Algorithms problems on LeetCode and CodeChef.
- Attained a 1600+ contest rating and maintained 200-day coding streak.

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI, Coursera
- Generative AI and LLMs: Architecture and Data Preparation — IBM, Coursera
- OCI 2025 Certified AI Foundations Associate — Oracle